

Andreas Demosthenous

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Summary

CS undergraduate at the University of Cyprus (2023–2027) and Software Engineer intern at the KIOS Research and Innovation Center of Excellence. Work spans geospatial Python, full-stack web, embedded firmware, low-level Linux, self-hosted infrastructure, and on-device LLM inference.

Experience

Software Engineer Intern, KIOS Research and Innovation Center of Excellence – Nicosia, Cyprus Jan 2025 – present

- Develop Python / QGIS plugins for geospatial water-network modelling; design data-ingestion pipelines turning tabular utility datasets into topologically correct GIS layers.
- Set up CI/CD for in-house plugin distribution using GitHub Actions and Git LFS.

Education

University of Cyprus, BSc in Computer Science – Nicosia, Cyprus Sept 2023 – June 2027

- Coursework: Operating Systems, Computer Architecture, Distributed Systems, Algorithms, Cryptography, Software Engineering.

Projects

Self-hosted homelab (Docker, ~25 services)

- Run a 25-service Docker Compose stack (Nextcloud, Vaultwarden, Paperless-ngx, Jellyfin, Gitea, Home Assistant) behind Traefik with Let's Encrypt and a Cloudflare Tunnel; migrated it from a Raspberry Pi 5 to an x86 QuickSync NAS.
- Ran a security audit and hardened the stack (socket-proxied Watchtower, no-new-privileges, reverse-proxy auth, secret rotation, key-only SSH, restic encrypted backups), then segmented a two-site UniFi network with VLANs, a zone-based firewall, and a WireGuard site-to-site mesh.

baseplate (open source)

- Opinionated single-host self-hosting template with hardened defaults (socket-proxied Watchtower, no-new-privileges, security headers), Traefik, and a Cloudflare Tunnel. github.com/AndrewDemsDS/baseplate

Smart-home automation and custom security system

- Multi-node ESP32 / Raspberry Pi system integrating sensors, actuators, and a custom security layer; full-stack web UI for monitoring and control.

Android kernel and device-tree ports

- Downstream device trees and kernel forks for Xiaomi (lavender, sdm660-common) and Samsung (universal7880) on AOSP / LineageOS.

ollama-flake

- Nix flake packaging Ollama for reproducible local LLM inference (github.com/AndrewDemsDS/ollama-flake).

dotfiles (open source)

- Hyprland + Quickshell desktop (fork of illogical-impulse) with custom QML widgets I built: Home Assistant control, a VPN/network status pill, an RSS reader with article view, and a homelab glance panel wired to self-hosted services. github.com/AndrewDemsDS/dotfiles

Skills

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, Bash

Web & data: Full-stack development, PostgreSQL, REST APIs

Geospatial: PyQGIS, QGIS plugin development, EPANET tooling, pandas/openpyxl

Systems: Linux (Gentoo, NixOS, Arch), Android device porting, systemd, kernel building

AI / LLMs: Self-hosted inference (Ollama, llama.cpp), MCP server development

Infrastructure: Docker / Compose, Traefik, Cloudflare Tunnels, restic, Nix

Networking: VLAN segmentation, UniFi, zone-based firewalls, WireGuard, site-to-site VPN

Hardware: ESP32, Raspberry Pi, Arduino

Spoken: Greek (native), English (B2+, English for Computer Science certification)